Program 09

Organize kubernetes Pods using labels and namespace

Structure of Program:

namespaces.yaml
pods.yaml

Steps:

1. Create two files namespaces.yaml and pods.yaml

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ nano namespaces.yaml
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ nano pods.yaml
```

2. Write the namespaces.yaml file

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
---
apiVersion: v1
kind: Namespace
metadata:
  name: prod
```

3. Verify the pods with namespace

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
```

```yaml
  containers:
    - name: frontend
      image: nginx
      ports:
        - containerPort: 80
---
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
    - name: backend
      image: nginx
      ports:
        - containerPort: 80
```

4. Apply the files

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
```

5. Verify the pods with namespace

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl get pods -n dev
NAME           READY   STATUS         RESTARTS   AGE
backend-pod    0/1     ErrImagePull   0          12s
frontend-pod   0/1     ErrImagePull   0          12s
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl get pods -n prod
NAME           READY   STATUS             RESTARTS   AGE
backend-pod    0/1     ImagePullBackOff   0          21s
frontend-pod   0/1     ImagePullBackOff   0          21s
```

6. Verify the pods

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl get pods
NAME                                READY   STATUS      RESTARTS   AGE
mycron-29435315-pnbrk               0/1     Completed   0          11m
mycron-29435320-c4ggv               0/1     Completed   0          6m26s
mycron-29435325-4br58               0/1     Completed   0          86s
nginx-deployment-6f9664446b-dfglq   1/1     Running     0          31m
nginx-deployment-6f9664446b-hn7s7   1/1     Running     0          31m
```

7. Filter the pods with labels

Verify the frontend pod with dev namespace

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl get pods -n dev -l app=frontend
NAME           READY   STATUS             RESTARTS   AGE
frontend-pod   0/1     ImagePullBackOff   0          5h34m
```

Verify the backend pod with prod namespace

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl get pods -n prod -l app=backend
NAME          READY   STATUS             RESTARTS   AGE
backend-pod   0/1     ImagePullBackOff   0          3m52s
```

8. Delete the pods

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl delete pods --all
pod "mycron-29435315-pnbrk" deleted from default namespace
pod "mycron-29435320-c4ggv" deleted from default namespace
pod "mycron-29435325-4br58" deleted from default namespace
pod "nginx-deployment-6f9664446b-dfglq" deleted from default namespace
pod "nginx-deployment-6f9664446b-hn7s7" deleted from default namespace
```

9. Delete the services

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
service "nginx-service" deleted from default namespace
```

10. Delete the deployments

```
_1RV24MC019_anusha_s@anu-Vivobook-Go-E1504FA-E1504FAB:~/pro-9$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted from default namespace
```